

Suman Panta

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Professional Summary

System and AI Administrator with 5+ years of enterprise IT experience and an M.S. in Information Technology, spanning Identity and Access Management, Endpoint and Network Operations, Cloud infrastructure, and Cybersecurity. Led enterprise AI administration for Microsoft 365 Copilot and Claude Enterprise, owning SSO/SAML federation, SCIM provisioning, license and access governance, DLP controls for AI interactions, and organization-wide secure AI adoption. Implemented Zero Trust identity controls in Microsoft Entra ID, enforced data governance through Microsoft Purview and Netwrix, and hardened endpoints with Microsoft Defender XDR and Intune in support of HIPAA, PCI DSS, and NIST CSF compliance. Automated infrastructure and compliance operations using Python, PowerShell, and the Microsoft Graph API, and engineered GenAI systems spanning RAG pipelines, MCP servers, and agentic workflows.

Skills & Technologies

AI Administration & Governance: Microsoft 365 Copilot Admin, Claude Enterprise Admin, AI Governance & Policy, Responsible AI, Enterprise AI Adoption, AI Risk Management, Purview AI Hub, Agent & Connector Governance, Model Context Protocol (MCP), AI Usage Analytics

Identity & Access Management: Microsoft Entra ID, Conditional Access, Privileged Identity Management (PIM), SSO/SAML, OAuth 2.0, SCIM Provisioning, MFA, Access Reviews, Zero Trust, Active Directory (AD DS), Entra Connect

Security & Compliance: Microsoft Defender XDR, Microsoft Purview, Data Loss Prevention (DLP), Sensitivity Labels, Insider Risk Management, eDiscovery, SIEM/Sentinel, KQL Threat Hunting, Incident Response, Vulnerability Management, NIST/CIS Frameworks

Systems & Endpoint: Linux (RHEL/Ubuntu), Windows Server (DNS, DHCP, GPO), macOS, Microsoft 365, Intune, Autopilot, Exchange Online, SharePoint, OneDrive, Teams, VMware/Hyper-V, Patch Management

Networking: TCP/IP, DNS, DHCP, VPN, VLANs, Firewalls, Routing & Switching, Wireshark, Network Troubleshooting

Languages & Scripting: Python, PowerShell, Bash, JavaScript (ES6+), TypeScript, SQL, Java

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, AI Agents, Vector Embeddings, OpenAI API, Anthropic Claude, Google Gemini, Mistral AI, Hugging Face Transformers

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Feature Engineering, Model Training & Fine-Tuning, Model Evaluation, Data Preprocessing

Backend & APIs: FastAPI, Flask, Django, Node.js, Express.js, REST APIs, WebSockets, Microsoft Graph API

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap

Databases: ChromaDB, Pinecone, MySQL, MongoDB, Oracle DB, Firebase

Cloud & DevOps: Microsoft Azure, AWS (EC2, S3, CloudWatch), Docker, Terraform, Jenkins, Git/GitHub, CI/CD

Work Experience

Pine Street Inn

Feb 2022 – Present

System Administrator

Boston, MA

- Led enterprise AI administration for Microsoft Copilot and Claude Enterprise, overseeing identity integrations (SSO/SAML and SCIM), license governance, AI security controls, and organization-wide AI adoption initiatives.
- Administered Microsoft Purview solutions including DLP, Information Protection, Sensitivity Labels, Insider Risk Management, and Compliance Manager to safeguard sensitive data across Microsoft 365 and AI platforms.
- Managed identity and access governance through Microsoft Entra ID, Conditional Access, MFA, RBAC, and privileged access controls to strengthen the organization's security posture.
- Collaborated with Security and Network teams to implement Zero Trust principles, perform threat remediation, and support incident response activities using Microsoft Defender XDR and Netwrix, maintaining compliance with HIPAA, PCI DSS, and NIST security frameworks.
- Configured and supported network infrastructure alongside the Network team, including firewalls, managed switches, wireless access points, and VPN, while maintaining VLAN segmentation, DNS/DHCP, and network access control (NAC) to isolate sensitive systems and monitor performance.
- Maintained Microsoft 365, Intune, Exchange Online, SharePoint Online, Teams, Windows Server, and cloud services while automating operational processes using PowerShell and Python.

Helpdesk Specialist

Boston, MA

- Delivered Tier 1–3 technical support for software, security, operating systems, and network connectivity across Windows and Linux environments.
- Developed Python and PowerShell automation scripts to streamline administrative workflows, reducing repetitive operational tasks.
- Administered enterprise identity and endpoint management using Active Directory, Microsoft Entra ID, Microsoft 365, Group Policy, and Intune across Windows environments.
- Supported infrastructure modernization initiatives including operating system migrations, endpoint security, cloud service adoption, and enterprise device lifecycle management.

- Troubleshoot complex software, networking, authentication, and endpoint issues while maintaining SLA compliance for enterprise users.

Eisai G2D2

Jan 2021 – Apr 2021

Junior System Administrator Intern

Cambridge, MA

- Provisioned and maintained AWS EC2 instances and supported cloud-based infrastructure operations.
- Automated administrative tasks using shell scripts; monitored Data Guard reports and system logs.
- Built, containerized, and deployed applications using Docker on EC2; gained hands-on exposure to Kubernetes and DevOps tools.
- Monitored system health using CloudWatch and researched monitoring and log aggregation tools including Datadog and Grafana.
- Developed Bash and Python scripts to streamline deployment, backup, and log analysis processes, improving operational efficiency.
- Assisted in optimizing cloud resource usage and implemented automated workflows for cloud infrastructure management.

Projects

GitHub MCP Server – Enterprise AI Tool Integration | Python, MCP SDK, PyGithub, REST APIs Sept 2026

- Built a Model Context Protocol (MCP) server enabling Claude Desktop to securely interact with GitHub through exposed API tools, extending LLM capabilities beyond static context into live repository operations.
- Implemented token-based authentication with GitHub Personal Access Tokens and isolated credential management via environment configuration, enforcing least-privilege scopes and preventing secret exposure in source control.
- Designed a layered architecture (Claude Desktop → MCP Server → GitHub API Client → GitHub REST API) with modular repository management tools, supporting extensibility for additional endpoints and enterprise AI agent workflows.
- GitHub Repository

StudyMate AI – RAG Powered Document Assistant | Python, LangChain, Mistral AI, ChromaDB, FastAPI Aug 2026

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) application enabling conversational question answering over custom knowledge bases using Large Language Models (LLMs).
- Designed a modular ingestion pipeline supporting PDF, DOCX, TXT, and web content with automated document parsing, chunking, embedding generation, and vector indexing in ChromaDB.
- Implemented semantic retrieval using vector embeddings and conversational memory with LangChain to improve context relevance and reduce hallucinations during LLM inference.
- Architected reusable retrieval, embedding, and document processing modules for scalable knowledge ingestion and future integration with additional embedding models and vector databases.
- GitHub Repository

Car Price Prediction using Linear Regression | Python, Scikit-learn, Pandas, NumPy Jul 2026

- Built an end-to-end supervised machine learning pipeline to predict used vehicle prices from structured automotive datasets using Linear Regression.
- Performed data preprocessing, feature engineering, categorical encoding, and feature selection to improve model quality and training performance.
- Evaluated model performance using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score while validating prediction accuracy on unseen test data.
- Conducted exploratory data analysis and feature visualization to identify relationships between vehicle attributes and selling price.
- GitHub Repository

GitHub Agentic Assistant | React.js, Node.js, Claude Code, REST APIs Feb 2026

- Developed an AI-powered GitHub assistant that streamlines repository interactions through an intelligent conversational interface.
- Integrated a React frontend with a Node.js backend to orchestrate AI-assisted workflows for repository management and developer productivity.
- Leveraged Claude Code to accelerate feature development, debugging, and code generation using agentic software engineering workflows.
- Designed a modular full-stack architecture enabling future integration with GitHub APIs, LLM providers, and autonomous coding agents.
- GitHub Repository

AI for Cyber Threat Detection | Python, TensorFlow, Scikit-learn, Data Engineering Jan 2026

- Co-authored a peer-reviewed research paper proposing a machine learning framework for anomaly detection across government and enterprise network environments.
- Designed data preprocessing and feature engineering pipelines to improve the effectiveness of supervised threat detection models.
- Evaluated detection performance using precision, recall, and anomaly detection metrics to assess model effectiveness on cybersecurity datasets.
- Research Publication

Sentiment Analysis on Product Reviews | Python ML, React.js, Node.js, Naive Bayes

Jun 2025

- Designed and deployed a full-stack sentiment analysis platform integrating React frontend with Node.js REST APIs
- Trained and evaluated a Naive Bayes classifier using scikit-learn and CountVectorizer on real-world datasets
- Achieved accurate sentiment classification and enabled real-time prediction through API endpoints
- GitHub Repository

Spam Email Detector Application | Python ML, React.js, Node.js

May 2025

- Developed an end-to-end spam detection system leveraging Naive Bayes and RESTful API integration
- Implemented text preprocessing, feature extraction, and model deployment using scikit-learn
- Integrated the ML model with a React frontend to enable interactive user interface and real-time monitoring of spam predictions
- GitHub Repository

Education

University of the Cumberlands

Dec 2025

Master of Science in Information Technology

Williamsburg, KY

- GPA: 3.9/4.00, Dean's List
- Relevant Coursework: Cloud Computing, Generative AI, Data Analysis and Visualization, Data Science, Machine Learning, Business Intelligence, Software Engineering Principles, Object-Oriented Programming

Wentworth Institute of Technology

Dec 2021

Bachelor of Science in Computer Networking

Boston, MA

- GPA: 3.6/4.00, Dean's List
- Relevant Coursework: Databases, Web Design, Data Structures and Algorithms, Networking, Operating Systems, Statistics, Network and Linux Administration, Core Java, Network Security, Computer Architecture, System Design, Web Applications, Ethical Hacking, Project Management

Licenses & Certifications

Applied Machine Learning: Algorithms · LinkedIn Learning · Feb 2026

Certificate of Completion: Claude 101 · Anthropic · Feb 2026 · Credential ID foc98fqtkrp6

Hands-On AI: Build an Autonomous Agent with the Claude Agent SDK · LinkedIn Learning · Jan 2026

Academy Accreditation – Generative AI Fundamentals · Databricks · Jan 2026 · Credential ID 172189702

Leadership

Wentworth Institute of Technology

Sep 2020 – Sep 2021

Orientation Leader

Boston, MA

- Led and collaborated with a team of orientation leaders to mentor and guide incoming freshmen during Wentworth Opening Week